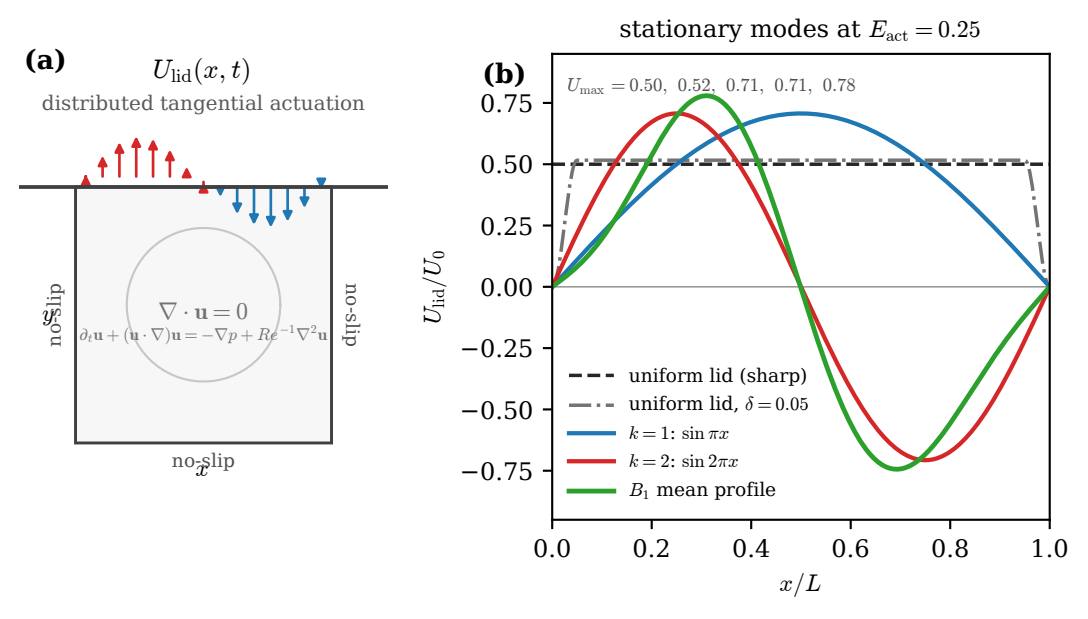
(a)
$U_{\mathrm{lid}}(x,t)$
distributed tangential actuation
no-slip
no-slip
$\nabla \cdot \mathbf{u} = 0$
$\partial_t \mathbf{u} + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\nabla p + Re^{-1}\nabla^2\mathbf{u}$
no-slip
y
x
stationary modes at $E_{\mathrm{act}} = 0.25$
(b)
$U_{\max} = 0.50,\ 0.52,\ 0.71,\ 0.71,\ 0.78$
U_{lid}/U_0
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
uniform lid (sharp)
uniform lid, $\delta = 0.05$
$k = 1$: $\sin \pi x$
$k = 2$: $\sin 2\pi x$
B_1 mean profile
0.0
0.2
0.4
0.6
0.8
1.0
x/L